Control Engineering Fundamentals

اسس هندسة السيطرة

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Control Engineering Fundamentals

Chapter 1

Introduction to Control System

1.1 Introduction

The automatic control system played a vital role of the development process in the engineering & science fields. It is a very important part of any system, from the smallest to the biggest systems. In addition to its extreme importance in high technology fields like space vehicles, missiles guidance, aircrafts...etc. Therefore, most of engineers & scientists must have a good understanding for the concept of control.

System:

present challenge to control engineers is the modeling and control of modern, complex, interrelated systems such as traffic control systems, chemical processes, and robotic systems. A system is a collection of components which interact with each other and with the environment (by information or energy links) from which the system is separated by a notational boundary.

A control system is an interconnection of components forming a system configuration that will provide a desired system response.

1.2 Control System

What is a control system? A control system is an interconnection of components forming a system configuration that will provide a desired system response.

The Objective of control system is to make the system OUTPUT and the desired REFERENCE as close as possible, i.e., to make the ERROR as small as possible.

Typically, the output variable is:

- Temperature in thermal systems.
- Position and velocity in mechanical systems.
- Voltage, current, charge, or frequency in electrical systems

benefits of an effective control system in production and manufacturing

There are many advantages offered to owners and employees of a production company when making use of a tailored control system that meets their company needs.

- Quality assurance
- Less wastage
- Lower operating costs
- Production planning
- Production control.

1.3 Classification of Control Systems

Based on some parameters, we can classify the control systems into the following ways.

1.3.1 SISO and MIMO Control Systems

- Control Systems can be classified as SISO control systems and MIMO control systems based on the **number of inputs and outputs** present.
- **SISO** (Single Input and Single Output) control systems have one input and one output. Whereas **MIMO** (Multiple Inputs and Multiple Outputs) control systems have more than one input and more than one output.

1.3.2 Open Loop and Closed Loop Control Systems

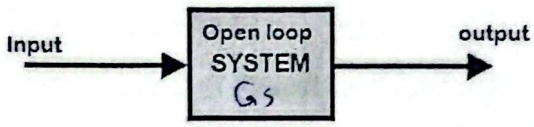
Control Systems can be classified as open loop control systems and closed loop control systems based on the **feedback path**.

In **open loop control systems**, output is not fed-back to the input. So, the control action is independent of the desired output.

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1) Open loop control system:

Any physical system which does not automatically correct the variation in its output is called an **open loop system** or *control system in which the output quantity has no effect upon the input quantity are called* **open loop control system**. This means that the output is not feedback to the input for correction.



In open loop control system, the output can be varied by varying the input. But due to external disturbances, the system output may change. When the output changes due to disturbances, it is not followed by changes in input to correct the output. In **open loop systems**, the changes in output are corrected by changing the input manually.

Advantages of Open loop control system:

1. The **open loop systems** are *simple* and economical.
2. The open loop systems are *easier* to construct.
3. Generally, the **open loop systems** are *stable*.

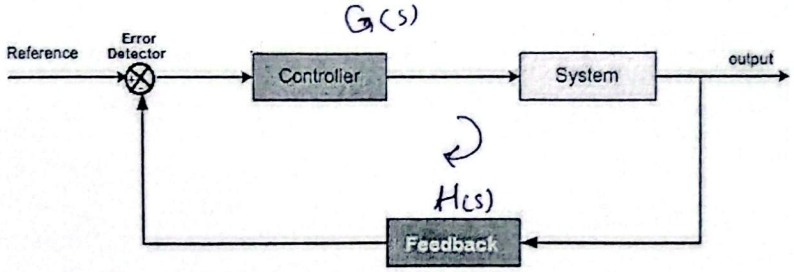
Disadvantages of open loop systems:

1. The **open loop systems** are *inaccurate and unreliable*.
2. The changes in the output due to external disturbances are not corrected automatically.

2) Closed loop control system:

Control systems in which the output has an effect upon the input quantity in order to maintain the desired output value are called **closed loop systems**.

The **open** loop **system** can be modified as closed loop system by providing a feedback. The provision of feedback automatically corrects the changes in output due to disturbances. Hence the closed loop system is also called automatic control system. The general **block diagram of an automatic control system** is shown in the figure below. It consists of an error detector, a controller, plant (open loop system) and feedback path elements.



The reference signal (or input signal) corresponds to the desired output. The feedback path elements sample the output and convert it to the same type as that of the reference signal. The feedback signal is proportional to the output signal and it is fed to the error detector. The error signal generated by the error detector is the difference between the reference signal and the feedback signal. The controller modifies and amplifies the error signal to produce better control action. The modified error signal is fed to the plant to correct its output.

Advantages of closed loop systems:

1. The **closed loop systems** are ***accurate***.
2. The ***sensitivity*** of the systems may be made small to make the system more stable.
3. The closed loop systems are ***less affected by noise***.

Disadvantages of closed loop systems:

1. The closed loop systems are ***complex and costly***.
2. The feedback in closed loop system ***may lead to an oscillatory response***.

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The differences between the open loop and the closed loop control systems are mentioned in the following table.

Open Loop Control Systems	Closed Loop Control Systems
Control action is independent of the desired output.	Control action is dependent of the desired output.
Feedback path is not present.	Feedback path is present.
Easy to design.	Difficult to design.
These are economical.	These are costlier.
Inaccurate. unrealable	Accurate. realable

1.4 definition of terms

Transfer Function T. F

frequency domain : تتعامل

In control theory, functions called transfer functions are commonly used to characterize the *input-output relationships* of components or systems that can be described by linear, time-invariant, *differential equations*.

frequency domain يعني

What is a Transfer Function?

The transfer function of a control system is defined as *the ratio of the Laplace transform of the output variable to Laplace transform of the input variable assuming* ***all initial conditions to be zero.***

$$TF = \frac{output}{input} = \frac{C(s)}{R(s)}$$

Disturbance: a signal which tends to adversely affect the value of the output of a system. There're two kinds of disturbance:

a) **Internal Disturbance** / generated within the system.

b) **External Disturbance** / generated outside the system and considered as an input.

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Servomechanism: It is a feed back control system in which the output is a mechanical position, velocity or acceleration.

Automatic Regulating System: It is a feedback control system in which the reference input, or the desired output, is either constant or slowly varying with the time, where the primary task is to maintain the actual output in the presence of disturbance. For example, the thermostat in the refrigerator motor.

1.5 Mathematical Modeling of Dynamic Systems

In studying control systems, we must be able to model dynamic systems and analyze dynamic characteristics. ***A mathematical model of a dynamic system*** *is defined as a set of equations that represents the dynamics of the system accurately* or, at least, well. Note that a mathematical model is not unique to a given system. A system may be represented in many ways and, therefore, may have many mathematical models, depending on one's perspective.

The dynamics of many systems, whether they are *mechanical, electrical, thermal, economic, biological*, and so on, may be described in terms of *differential equations*. Such differential equations may be obtained by using physical laws governing a particular system, for example, Newton's laws for mechanical systems and Kirchhoff's laws for electrical systems. We must always keep in mind that deriving reasonable mathematical models is the most important part of the entire analysis of control systems.

1.6 Mathematical Model of Physical System

To understand and control complex system, we must find the mathematical models of these systems. The term ***mathematical model***, in control engineering perspective, implies a set of differential equation that describes the dynamic behavior of a process.

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1.6.1 Mathematical Model of Electrical System

Consider RLC circuit as shown in fig below

It consists of:

Current Source $i(t)$

Resistor characterized by its resistance R

Inductor characterized by its inductance L

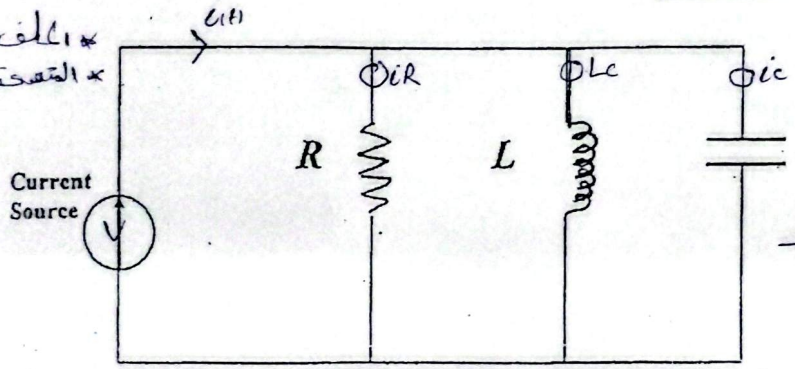
Capacitor characterized by its capacitance C

$Z = R + j\omega$
$Z = R + j\omega L$
$Z = R - j\omega C$

Differential equations of the RLC circuit can be obtained by utilizing *Kirchhoff's* laws and the voltage-current relationships for R, L & C.

$R = \frac{V}{i}$, $V = iR$, $i = \frac{V}{R}$

* المكثف يخزن تيار
* الملفة تخزن فولتية



$C \quad v = L \frac{di}{dt}$, $i = \frac{1}{L}\int v$

$i = C \frac{dv}{dt}$, $v = \frac{1}{C}\int i\,dt$

$i(t) = i_r + i_L + i_C$

$i(t) = \frac{V}{R} + \frac{1}{L}\int v\,dt + C\frac{dv}{dt}$

So, according to *Kirchhoff's* current law:

$i(t) = i_r + i_l + i_c$

Where i_r , i_l & i_c are currents through the resistor, inductor & capacitor respectively.

On the other hand ***Kirchhoff's*** voltage law implies that the voltage across any of the mentioned elements is the same.

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